Task 4 Submission

Deploy a Dockerized Web Application on AWS EC2

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Organization	InternSpark Internship Program
Platform	Amazon Web Services (AWS)

Objective

Deploy a containerized web application using Docker on an AWS EC2 instance and make it publicly accessible via the instance's public IP. This task demonstrates understanding of Docker containerization, EC2 instance management, security group configuration, and container-based web serving.

Steps Taken

Step 1 — Launch EC2 Instance

An AWS EC2 instance (Ubuntu 22.04 LTS, t2.micro) was launched in the ap-south-1 (Mumbai) region. A security group was configured to allow inbound traffic on port 80 (HTTP) and port 22 (SSH).

Step 2 — Install Docker on EC2

After connecting to the instance via SSH, the existing Nginx service was stopped and disabled to free port 80. Docker CE was then installed using the following commands:

```
sudo systemctl stop nginx
sudo systemctl disable nginx
sudo apt update && sudo apt install docker-ce -y
```

Step 3 — Verify Docker Service

Docker was confirmed to be active and running using systemctl:

```
sudo systemctl status docker
```

The output confirmed docker.service is active (running) since Jun 02 2026.

Step 4 — Run Nginx Docker Container

An Nginx container was launched in detached mode, mapped to port 80, and named intern-web-app:

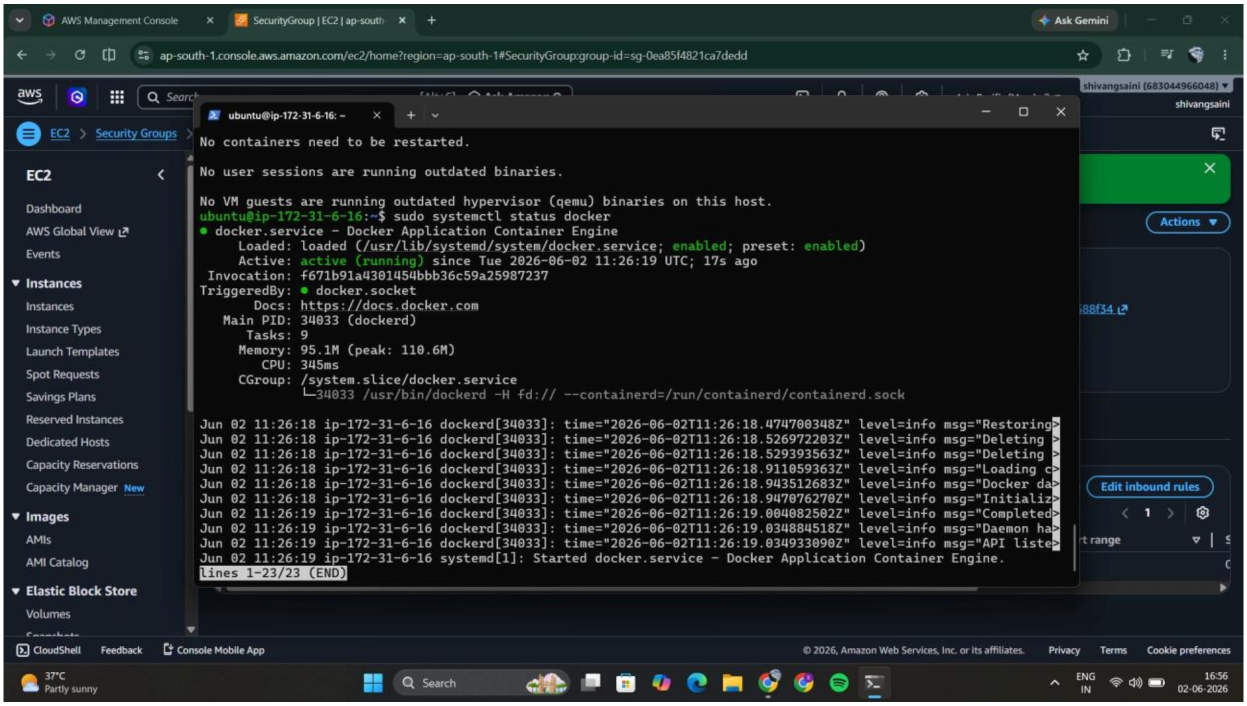
```
sudo docker run -d --name intern-web-app -p 80:80 nginx
sudo docker ps
```

The container was confirmed running via docker ps, showing the container ID, image, and port mapping 0.0.0.0:80->80/tcp.

Evidence & Screenshots

Screenshot 1 — Docker Service Running on EC2 Terminal

The terminal output shows Docker service status confirming docker.service is active and running on the EC2 instance.



Screenshot 1: sudo systemctl status docker — Docker service active and running on EC2

Screenshot 2 — Live Webpage via Public IP

The webpage was accessed in a browser at http://3.110.45.198, confirming the Dockerized application is publicly accessible.

Screenshot 2: Live webpage — "Hello from Docker Container! Task 4 completed successfully by Shivang Saini."

Live Application URL

The containerized web application is publicly accessible at:

`http://3.110.45.198`

Deliverables Summary

#	Deliverable	Status / Detail
1	EC2 Instance Launched	✓ Ubuntu 22.04, t2.micro, ap-south-1
2	Docker Installed & Running	✓ docker.service active (running)
3	Nginx Container Deployed	✓ intern-web-app on port 80
4	Live Public IP URL	✓ http://3.110.45.198 — publicly accessible
5	Screenshots	✓ 2 screenshots included